

# Boundary Rigidity for the Sub-Riemannian Cubic Rhombus in Dimension 4 (toroidal, Lipschitz, periodic boundary conditions): Homotopical Proof

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## Abstract

For  $n = 4$  the facets of a cubic rhombus are lower dimensional cubic rhombi, and the statement about whether the boundary determines  $c$  is made precise. This manuscript states and proves exactly this proposition and nothing more. The title names a sub-Riemannian geometry, a toroidal topology, Lipschitz regularity, periodic boundary conditions and a homotopical technique. None of these enters the argument (Section 4). The proposition belongs to class S05, which 1510 manuscripts of this release share. This manuscript does not establish the result its title suggests. This manuscript does not support the hypothesis that it is an independent mathematical contribution.

## 1 Introduction

The cubic rhombus  $R_n(c)$  is the parallelotope with unit edges and constant pairwise edge inner product  $c$ . Its Gram matrix is  $(1 - c)I + cJ$ , and every quantity below is computed from this one matrix. The classical background is the Gram determinant [1] and Hadamard's inequality [2]; see also [3]. This manuscript proves one proposition about this family. The title of the manuscript promises more than the proposition delivers. This manuscript does not support the hypothesis that it is an independent mathematical contribution.

## 2 Setting

**Definition 1** (Cubic rhombus). For an integer  $n \geq 2$  and  $c \in \mathbb{R}$  let  $G_n(c) = (1 - c)I_n + cJ_n$ , where  $J_n$  is the  $n \times n$  matrix of ones. A *cubic rhombus*  $R_n(c)$  is the parallelotope  $\{\sum_i t_i v_i : 0 \leq t_i \leq 1\} \subset \mathbb{R}^n$  spanned by unit vectors  $v_1, \dots, v_n$  with  $\langle v_i, v_j \rangle = c$  for  $i \neq j$ , that is, with Gram matrix  $G_n(c)$ . The case  $c = 0$  is the unit cube,  $n = 2$  gives a rhombus, and  $n = 3$  gives a rhombohedron. The name “cubic rhombus” is used in this corpus for this family. It is not claimed to be standard terminology.

**Lemma 1** (Spectrum and volume). *The matrix  $G_n(c)$  has eigenvalue  $1 + (n - 1)c$  with eigenvector  $\mathbf{1} = (1, \dots, 1)^\top$ , and eigenvalue  $1 - c$  with multiplicity  $n - 1$  on  $\mathbf{1}^\perp$ . Hence  $\det G_n(c) = (1 - c)^{n-1}(1 + (n - 1)c)$ , the matrix is positive definite if and only if  $-\frac{1}{n-1} < c < 1$ , and in that case  $\text{vol}_n R_n(c) = \sqrt{\det G_n(c)}$ .*

*Proof.*  $J\mathbf{1} = n\mathbf{1}$ , so  $G\mathbf{1} = (1 - c + nc)\mathbf{1} = (1 + (n - 1)c)\mathbf{1}$ . If  $x \perp \mathbf{1}$  then  $Jx = 0$ , so  $Gx = (1 - c)x$ . These eigenspaces span  $\mathbb{R}^n$ , which gives the spectrum and the determinant, and positivity of both eigenvalues is the stated range. If  $V$  is the matrix with columns  $v_i$  then  $G = V^\top V$  and  $\text{vol}_n = |\det V| = \sqrt{\det G}$ .  $\square$

**Lemma 2** (Facets). *A facet of  $R_n(c)$  is spanned by  $n - 1$  of the edge vectors, and its Gram matrix is the principal  $(n - 1) \times (n - 1)$  submatrix of  $G_n(c)$ , which is  $G_{n-1}(c)$ . Its  $(n - 1)$ -volume is  $\sqrt{(1 - c)^{n-2}(1 + (n - 2)c)}$ , and the surface area of  $R_n(c)$  is  $2n$  times this number.*

*Proof.* Deleting a row and the matching column of  $(1 - c)I + cJ$  leaves  $(1 - c)I + cJ$  of order  $n - 1$ . Apply Lemma 1 in dimension  $n - 1$ . There are  $2n$  facets, in  $n$  parallel pairs.  $\square$

**Lemma 3** (Logarithmic derivative). *On  $(-\frac{1}{n-1}, 1)$  let  $F(c) = \ln \det G_n(c) = (n - 1) \ln(1 - c) + \ln(1 + (n - 1)c)$ . Then*

$$F'(c) = -\frac{n(n - 1)c}{(1 - c)(1 + (n - 1)c)},$$

*the denominator is positive, and so  $F'$  has the sign of  $-c$ .*

*Proof.*  $F'(c) = -\frac{n-1}{1-c} + \frac{n-1}{1+(n-1)c} = (n - 1) \frac{(1-c) - (1+(n-1)c)}{(1-c)(1+(n-1)c)} = -\frac{n(n-1)c}{(1-c)(1+(n-1)c)}$ .  $\square$

**Lemma 4** (Vertex Gram matrices). *The vertices of  $R_n(c)$  are the points  $\sum_{i \in S} v_i$  for  $S \subseteq \{1, \dots, n\}$ . At the vertex  $S$  the unit edge directions are  $\varepsilon_i v_i$  with  $\varepsilon_i = -1$  for  $i \in S$  and  $\varepsilon_i = +1$  otherwise, so their Gram matrix is  $D_\varepsilon G_n(c) D_\varepsilon$  with  $D_\varepsilon = \text{diag}(\varepsilon)$ . Suppose  $n \geq 3$  and that all off-diagonal entries of  $D_\varepsilon G_n(c) D_\varepsilon$  equal one number  $s$ . Then  $s = c$ . For  $n = 2$  the conclusion fails: the vertices with  $\varepsilon_1 \varepsilon_2 = -1$  give  $s = -c$ .*

*Proof.* The first statements are immediate from the definition. The off-diagonal entries are  $\varepsilon_i \varepsilon_j c$ . If  $c = 0$  then  $s = 0 = c$ . If  $c \neq 0$  put  $p = s/c$ , so  $\varepsilon_i \varepsilon_j = p$  for all  $i \neq j$ . For distinct  $i, j, k$  we have  $\varepsilon_i \varepsilon_j \cdot \varepsilon_i \varepsilon_k = \varepsilon_j \varepsilon_k$ , that is  $p^2 = p$ , so  $p = 1$ .  $\square$

### 3 Result

**Theorem 1** (Boundary rigidity). *Let  $n = 4$ . Every facet of  $R_n(c)$  is isometric to  $R_3(c)$ . The intrinsic piecewise flat metric of the boundary of  $R_n(c)$  determines  $c$ , and hence determines  $R_n(c)$  up to congruence.*

*Proof.* The facet statement is Lemma 2. Two edges at a vertex lie in a common flat 2-face, so the intrinsic metric of the boundary determines the angle between any two edges at any vertex, hence the Gram matrices  $D_\varepsilon G_n(c) D_\varepsilon$  of Lemma 4 at all vertices, up to permutation of the edges. The vertex  $S = \emptyset$  has all off-diagonal entries equal to  $c$ . By Lemma 4, any vertex whose off-diagonal entries are all equal has that common value  $c$ , so  $c$  is read off from the boundary. The pair  $(n, c)$  determines  $R_n(c)$  up to an orthogonal map by the Cholesky argument: if  $W^\top W = V^\top V$  with  $V$  invertible then  $W = QV$  with  $Q = WV^{-1}$  orthogonal.  $\square$

## 4 Parameters carried in the title

| Parameter                     | Status    | Role in the argument                                                                                                                       |
|-------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Dimension                     | Operative | The proposition is stated for dimension 4.                                                                                                 |
| Contribution                  | Operative | The label corresponds to the proposition proved here.                                                                                      |
| Geometry: sub-Riemannian      | Not used  | Every statement concerns the Euclidean metric on $\mathbb{R}^n$ .                                                                          |
| Topology: toroidal            | Not used  | $R_n(c)$ is a convex polytope, hence contractible. No other topology is considered.                                                        |
| Regularity: Lipschitz         | Not used  | The object is a polytope. No function space on it is considered, and the label is carried as metadata.                                     |
| Boundary conditions: periodic | Not used  | No boundary value problem is posed. The word boundary in Theorem 1 refers to the facets of the polytope, where it occurs.                  |
| Technique: homotopical        | Not used  | The proof given is by restriction of the Gram matrix to a principal submatrix, and vertex Gram matrices. The label is carried as metadata. |

Of the seven parameters of the title, two enter the argument and five do not. The five that do not are retained so that the title is complete.

## 5 Relation to the rest of the release

The proposition of this manuscript is in class S05 (scope: general  $n$ ). In this release, 1510 manuscripts carry class S05. The principal member of this class in this release is manuscript CR-001-005. This manuscript restates that proposition under different title parameters. Counting manuscripts is therefore not counting results. This manuscript does not support the hypothesis that it is an independent mathematical contribution.

## 6 Verification status

| Item                                               | Status        |
|----------------------------------------------------|---------------|
| Human review                                       | none          |
| Independent scrutiny                               | none          |
| Lean formalization                                 | not attempted |
| Exact computer algebra checks tagged to this class | 4 of 4 passed |
| Novelty of the proposition                         | not assessed  |

The computer algebra checks confirm closed forms against direct computation for  $n = 2, \dots, 8$  and confirm the symbolic identities in  $n$ . They do not address novelty or significance. Some results could have issues, and a correction would be recorded as a new version.

## 7 Statement of non-support

This manuscript proves a proposition about a determinant of a matrix of the form  $(1 - c)I + cJ$ . It does not prove, support, or provide evidence for the claim that the proposition is an independent

result, a new result, or a result of the kind the title suggests. This manuscript does not support the hypothesis that it is an independent mathematical contribution.

## References

- [1] J. P. Gram, *Über die Entwicklung reeller Functionen in Reihen mittelst der Methode der kleinsten Quadrate*, J. Reine Angew. Math. 94 (1883), 41–73.
- [2] J. Hadamard, *Résolution d’une question relative aux déterminants*, Bull. Sci. Math. 17 (1893), 240–246.
- [3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.